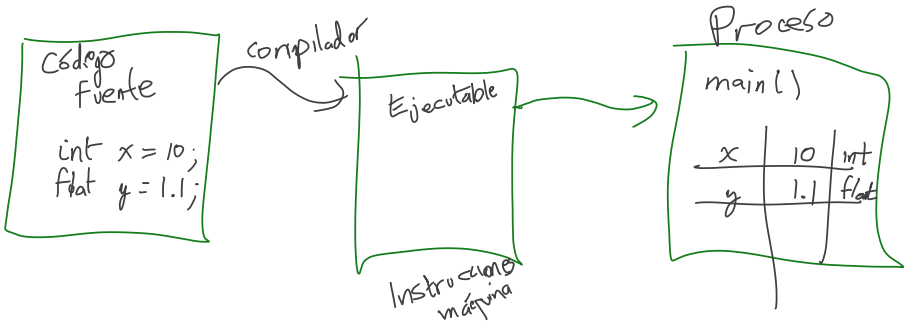
10 – Funciones Recursivas

miércoles, 25 de febrero de 2026 15:12

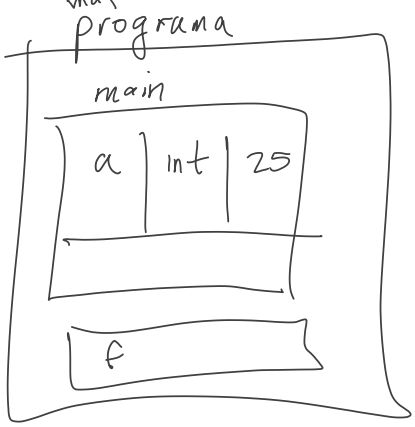
Procesos

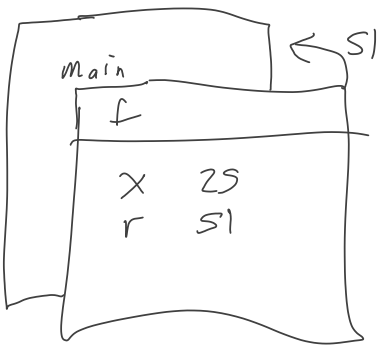



```
int main(){
    int a= 25;
    f(a);
}

int f(int x){
    int r= 2*x+1;
    return r;
}
```

Sistema Operativo





$$S = 1 + 2 + 3 + 4 + 5 + \dots + 100$$

$$S = \sum_{i=1}^{100} i = 100 + \sum_{i=1}^{99} i$$

$$= 100 + 99 + \sum_{i=1}^{98} i$$

$$\sum_{i=1}^{n} i = n + \sum_{i=1}^{n-1} i$$

Función Recursiva: Función que se llama a sí misma como parte de sus instrucciones

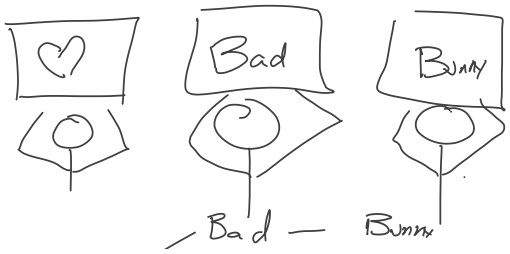
Partes de una función recursiva:

1- Llamada recursiva
2- Condición de paro

Ejercicio: Generar una función recursiva que calcule 3^n

$3^0 = 1$
$3^1 = 3$
$3^2 = 9$
$3^3 = 27$
$3^4 = 81 = 3 \cdot 3^3$
$3^5 = 243 = 3 \cdot 3^4$

Factorial



♡ ← Bunny — Bad

Bad ← Bunny — ♡

Bad ← ♡ ← Bunny

Bunny ← ♡ → Bad

Bunny ← Bad — ♡

$$3 \cdot 2 \cdot 1 = 6$$

$$5! = 5 \cdot 4!$$

$$5 \cdot 4 \cdot 3!$$

$$5 \cdot 4 \cdot 3 \cdot 2!$$

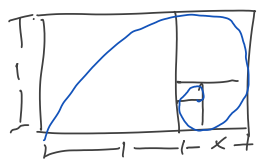
$$5 \cdot 4 \cdot 3 \cdot 2 \cdot 1!$$

$$5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$$

Fibonacci:

1 1 2 3 5 8 13 21 34 55 89 ...

1.5 1.66 1.6 1.625 1.6153 1.6190 1.6176 1.6181



$$\frac{1}{1+x} = \frac{x}{1}$$

1	1	2	3	5	8	13	21	34	55	89
1	2	3	4	5	6	7	8			

$$f(n) = f(n-1) + f(n-2)$$

$f(1) = 1$
$f(2) = 1$